Spring Semester Examination 2025
(Even Semester)
Aliah University

Course Title: Probability-II
Course Code: STAUGMCC1204
Time: 3 Hours **Full Marks: 80**

Instructions:
Answer all questions from **Section A** and **Section B**. Attempt any **six** questions from Section C. All questions carry marks as indicated.

Section A: Multiple Choice Questions $(2 \times 5 = 10)$

1. Which of the following distributions has memoryless property?
   (a) Normal (b) Binomial (c) Exponential (d) Hypergeometric
2. If $X \sim B(n, p)$, then $\text{Var}(X)$ is:
   (a) np (b) $np(1-p)$ (c) n^2p (d) np^2
3. Markov's inequality is applicable to which type of random variables?
   (a) Only positive random variables (b) Negative random variables (c) Zero mean random variables (d) Symmetric random variables
4. Which distribution is symmetric about zero?
   (a) Gamma (b) Beta (c) Normal (d) Poisson
5. Independence of two random variables implies:
   (a) Zero covariance (b) Equal variances (c) Equal expectations (d) Zero mean

Section B: Short Answer Questions $(2 \times 5 = 10)$

6. State two key assumptions for the application of the Central Limit Theorem (CLT).
7. What is meant by the "memoryless property" of a distribution? Name one distribution that satisfies it.
8. Define convergence in probability. How is it different from almost sure convergence?
9. Write down the condition for a function $\phi(x)$ to be applicable in Jensen's inequality.
10. Explain the significance of the covariance of two random variables being zero. Does it imply independence?

Section C: Attempt any Six $(10 \times 6 = 60)$

11. (a) Let $X \sim N(0, 1)$. Define $Y = X^2$. Find the distribution of Y. [3]
    (b) Discuss the transformation when $Y = \frac{1}{X^2}$ and $X \sim N(0, 1)$, and describe its type. [3]
    (c) Generalize the result: if $X \sim N(\mu, \sigma^2)$, determine the distribution of $Y = \frac{X-\mu}{\sigma}$. [4]
12. (a) Define the pmf of the Negative Binomial distribution and state its parameters. [3]
    (b) Define the pmf of the Hypergeometric distribution and explain the difference between Hypergeometric and Binomial distributions. [3]

(c) Derive the mean and variance of the Geometric distribution. [4]

13. (a) Define the probability density function (pdf) of the continuous uniform distribution over $[a, b]$. [2]

(b) Derive the mean and variance of the exponential distribution with rate parameter λ. [4]

(c) Compare the memoryless property of the exponential distribution with the uniform distribution. [4]

14. (a) State the pdf of the standard normal distribution and sketch its graph. [3]

(b) Define the lognormal distribution and explain how it is derived from a normal distribution. [3]

(c) Derive the expression for the mean of a lognormal distribution. [4]

15. Given the joint probability mass function of discrete random variables X and Y in the table below:

$Y\backslash X$	0	1	2	
0	0.1	0.1	0.1	→ 0.2
1	0.1	0.2	0.1	→ 0.4
2	0.1	0.1	0.1	→ 0.3
	0.3	0.4	0.2	

(a) Find the marginal distributions of X and Y. [2]

(b) Compute the conditional probability $P(Y = 1 \mid X = 1)$ and $P(X = 2 \mid Y = 0)$. [4]

(c) Check whether X and Y are independent. Justify your answer using joint and marginal probabilities. [4]

16. (a) Let the joint pdf of (X, Y) be

$$f(x, y) = \begin{cases} 6(1 - y), & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the marginal pdfs of X and Y. [3]

(b) Compute the expectations $E[X]$, $E[Y]$, and $E[XY]$. [3]

(c) Using the results above, compute $\text{Cov}(X, Y)$ and the correlation coefficient $\rho(X, Y)$. [4]

17. (a) State and prove Markov's inequality for a non-negative random variable. [3]

(b) State Chebyshev's inequality and prove it using Markov's inequality. [3]

(c) Let X be a random variable with $E(X) = 10$ and $\text{Var}(X) = 25$. Use Chebyshev's inequality to find an upper bound for $\mathbb{P}(|X - 10| \geq 10)$. [4]

18. (a) Define the following: convergence in probability, convergence in mean square. [2]

(b) Give an example of a sequence of random variables that converges in probability but not almost surely. [4]

(c) Explain the difference between convergence in distribution and convergence in probability. Prove that convergence in mean square implies convergence in probability. [4]

19. (a) State and prove the Central Limit Theorem (CLT) for i.i.d. random variables. [4]

(b) State the Weak Law of Large Numbers (WLLN) and its conditions. [2]

(c) Let $X_1, X_2, \ldots, X_{100}$ be i.i.d. random variables with $E(X_i) = 50$ and $\text{Var}(X_i) = 100$. Use CLT to approximate $\mathbb{P}(490 < \sum_{i=1}^{100} X_i < 510)$. [4]

End-Semester Examination - 2025 (Spring)
STAUGMDC1202
(Economics Statistics & Official Statistics)
Semester – II (4-years B.Sc. in Statistics)

Full Marks – 60 **Time – 2:30 Hours**

Notations and symbols have their usual meaning.

Section A. Answer **ALL** questions. Each question carries **Two** marks.

1. The principal source of the data related to Industrial Statistics in India is
   (a) Index of Industrial production (IIP)
   (b) Annual Survey of Industries (ASI)
   (c) Economic Census
   (d) Micro, Small and Medium Enterprises (MSME) Census
2. Which of the following statements is/are correct about average prices if the price index is 110?
   (a) The prices have increased by 10 per cent
   (b) The prices have increased by 110 per cent
   (c) The prices have decreased by 10 per cent
   (d) None of the above
3. The weights used in a quantity index are
   (a) Quantity
   (b) Values
   (c) Price
   (d) None of the above
4. Which of the following index has a downward bias?
   (a) Fisher's index
   (b) Marshall-Edgeworth index
   (c) Paasche's index
   (d) Lespeyre's index
5. If an index number calculations over 8 years with a base value of 100 gave an index for 2015 of 120, what would be the percentage relative for 2015?
   (a) 120
   (b) 12
   (c) 100
   (d) 200

Section B. Answer **ALL** questions. Each question carries **Two** marks.

6. What is a simple index number?
7. Define the consumer price index number.
8. What is meant by base shifting of the index number?
9. What is meant by deflating the index number?
10. Explain the time reversal test.

Section C. Answer any **Four** questions. Each question carries **Ten** marks.

11. (a) Prove that Fisher's ideal index number lies between Laspeyre's and Paasche's index numbers.
    (b) "Index numbers are economic barometers." Discuss it. [5+5]

12. Discuss the importance and limitations of index numbers. Explain the various steps in the construction of the wholesale price index numbers of a set of a group of 'related variables'. [2+2+6]

13. What does the cost of living index measure? How do you derive the weights required in constructing this index? State the important uses of this index. [2+5+3]

14. (a) Show that the Marshall-Edgeworth index lies between Laspeyre's and Paasche's index numbers.
    (b) Given that $\sum p_1q_1 = 250$, $\sum p_0q_0 = 150$, Paasche's index number $= 150$ and Drobish-Bowley's index number $= 145$. Find out (i) Marshal Edgeworth index number (ii) Fisher's index number. [5+5]

15. (a) What is a chain index? Explain the steps involved in the construction of chain indices.
    (b) What do you mean by circularity test? Show that the test is satisfied only by the indices based on the simple geometric mean of the price relative. [5+5]

16. (a) Outline the key roles and responsibilities of the National Sample Survey Organisation (NSSO).
    (b) Explain how official statistics support policy-making and governance. [5+5]

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End-Semester Examination - 2025 (Spring)
STATUGSEC1202
(Statistical Computations Using R)
Semester – II (4 Year B.Sc. in Statistics)
Time – 3 Hours
Full Marks: 50

Answer all the questions using R:

1. Suppose you have a vector x= (1,5,2,9,7,3,8,1,4,5,6,8,3,9,1)'. Find out the following:
   a. Find out the sum of the elements of this vector. 1
   b. Obtain another unit vector (with all the elements equal to unity) of the same dimension as x. Let that be y. Find out the sum of these two vectors. 2 + 2
   c. Find out the product of these two vectors. (both x'y and xy') 3
   d. Obtain the mean and sd of x. 2
2. Create a 4x4 matrix with the elements 1 to 16. Name it m1. Create another matrix with the elements 17 to 28 with 4 rows and 3 columns. Name it mat2. 4,4
   a. Add and multiply mat1 and mat 2 if compatible. 2
   b. Extract the second row from mat1 and first column from mat2. 2
   c. Add a constant 5 with mat1 1
   d. Calculate the cumulative sum ('running total') of the numbers 2,3,4,5,6.3. Again calculate the cumulative sum of those numbers, but in reverse order. 2
   e. Use the built in functions to find out the min and the max of the elements in mat1 and to find out the dimension of mat1. 3
3. The following lines of code contain some common errors that prevent them from being evaluated properly or result in error messages. Identify the errors and rectify them. 10

4,4 4x2

```
vector1 <- c('one', 'two', 'three', 'four, 'five', 'seven')

vec.var <- var(c(1, 3, 5, 3, 5, 1)

vec.mean <- mean(c(1, 3, 5, 3, 5, 1))
```

```
vec.Min <- Min(c(1, 3, 5, 3, 5, 1))

Vector2 <- c('a', 'b', 'f', 'g') vector2
```

4. a. Consider the rainfall data in R. rainfall <- c(0.1, 0.6, 59.1, 42.8, 1.9, 0.6, 3.6, 0.0, 61.7, 0.3, 0.0, 0.1, 0.0, 81.4, 61.7). Take a subset of the rainfall data where rain is larger than 20. 2
   b. Subset the vector where it is either exactly zero, or exactly 0.6. 2
   c. What is the 18th letter of the alphabet? 2
   d. Generate a random subset of fifteen letters. 2
   e. Find out whether there are any duplicate values? If yes, which ones? 2

5. a. Consider the mtcars data set in R and draw a histogram with "wt" and "mpg" and comment. 2
   b. Draw boxplots with "mpg" for different levels of "am" and comment. 2
   c. Draw the scatter plot of "wt" and "mpg" and comment. 1
   d. Construct a regression equation between the two and plot it and put the regression equation inside the plot. 2+1
   e. Predict the mpg of a car with weight 2.1 unit. 2

ALIAH UNIVERSITY
End-Semester Examination Spring-2025
Subject- Analytical Geometry and Matrix Algebra
Subject code: MATUGMCC1203/MATUGMIN1202.
Full marks-80.
Duration- 3Hrs.

Notations and symbols have their usual meaning.

Group A

Answer any ten questions $(10 \times 2 = 0)$

1. Define vector space.
2. Define rank of a matrix.
3. State Cayley-Hamilton theorem for a square matrix.
4. Define singular matrix with an example.
5. Define inverse of a matrix.
6. Define row-reduced echelon matrix.
7. Define elementary matrix.
8. Find the nature of the conic $9x^2 + 24xy + 16y^2 + 34x + 62y + 47 = 0$.
9. Find the equation of the tangent at the point $(1, 3)$ to the parabola $y^2 = 9x$.
10. Show that the equation $5x^2 - 6xy + y^2 = 0$ represents a pair of straight lines.
11. Find the equation of the chord of the circle $x^2 + y^2 = 81$ which is bisected at the point $(-2, 3)$.
12. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 5$ at the point $(2, 0, 1)$.

Group B

Answer any twelve questions $(12 \times 5 = 60)$

1. A non-empty subset W of a vector space V over a field F is a subspace of V if and only if $(i)\alpha + \beta \in W \ \forall \alpha, \beta \in W$ and $c\alpha \in W \ \forall \alpha \in W, \ \forall c \in F$. 5

2. (a) Let $M_n(\mathbf{R})$ be the vector space of $n \times n$ matrices over $\mathbf{R}$. Let U be the subset of $M_n(\mathbf{R})$ such that $U = \{A = (a_{ij}) : a_{22} + ... + a_{nn} = 0, a_{11} = 0\}$. Is it true that U is a subspace of V over $\mathbf{R}$? 3

(b) Prove that the set of all real valued continuous functions on the closed and bounded interval $[a, b]$ is a subspace of the vector space of all real valued functions on $[a, b]$. 2

3. (a) Prove that the set of all real polynomials of degree less or equal to n, $n \geq 1$ forms a vector space. 2

(b) Find the row space of the matrix $\begin{pmatrix} 0 & 3 & 7 \\ 2 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}$. 3

4. By converting to a row-reduced echelon matrix, obtain row rank of the following matrix 5

$$\begin{pmatrix} 1 & 2 & -1 \\ 7 & 4 & 3 \\ 2 & 2 & 1 \end{pmatrix}.$$

5. Find the eigenvalues and eigenvectors of the following matrix 5

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}.$$

6. If $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$, show that $A^2 - 10A + 16I_3 = O$. Hence obtain A^{-1}. 5

7. (a) Let A and B are real orthogonal matrices of the same order and $\det A + \det B = 0$. Show that $A + B$ is a singular matrix. 3

(b) Prove that a complex matrix can be uniquely expressed as the sum of a Hermitian matrix and a skew Hermitian matrix. 2

8. Use Cramer's rule to solve the following system of equations: 5

$$x + z = 0,\ 3x + 4y + 5z = 2,\ 2x + 3y + 4z = 1.$$

9. Determine the conditions for which the following system of equations $x+2y+z = 1$, $2x + y + 3z = b$, $x + ay + 3z = b + 1$ has

(a) only one solution, (b) no solution, (c) many solutions. 5

10. Reduce the equation $3x^2 + 2xy + 3y^2 - 16x + 20 = 0$ to canonical form and find the nature of the conic. 5

11. Find the set of angles by which the axes should be rotated so that the equation $x^2 - y^2 = 0$ becomes another equation in which xy term is present. 5

12. Find the condition in which the straight line $lx + my + n = 0$ touches the parabola $y^2 - 4px + 4pq = 0$. 5

13. Find the condition that the normals at the points (x_1, y_1), (x_2, y_2) and (x_3, y_3) on the ellipse $x^2/a^2 + y^2/b^2 = 1$ will be concurrent. 5 5

14. Find the equation of the polar of the point $(2, -1)$ with respect to the circle $2x^2 + 2y^2 - 3x + 5y - 7 = 0$. Find the pole of the straight line $9x + y - 29 = 0$ with respect to the same circle. 5 5

15. Show that the pair of straight lines whose direction cosines are given by $3lm - 4ln + mn = 0$ and $l + 2m + 3n = 0$ are at right angles. 5

16. Find the equation of the plane passing through the points $(2, 2, 1)$, $(3, 4, 2)$ and $(7, 0, -6)$. 5 5

17. Find the point where the straight line joining the points $(2, -3, 1)$ and $(3, -4, -5)$ cuts the plane $3x + y + z = 8$. 5 5

18. Find the equation of the sphere passing through the four points $(1, -1, -1)$, $(3, 3, 1)$, $(-2, 0, 5)$, $(-1, 4, 4)$. 5

$-\frac{3}{2}(x + x_1), m z_1$

UCCUGVAC1201 Environmental Science

Full marks-80 **Time-3 Hours**

Group - A

Give tick (√) any 20 questions from the following. 20 x 1 = 20

1.Minamata disease occurred due to
a. Mercury b. Carbon dioxide c. Sulphur dioxide

2.Name one water pollutants which causes the black foot disease
a. Sodium b. Arsenic c. Iron

3.What percentage of oxygen present in the atmosphere?
a.60% b. 21% c. 34%

4.How many National parks are present in India?
a. 45 b.100 c.106

5.Eutrophication is observed in
a. Atmosphere b. soil c. Hydrosphere

6.One example of micro consumer or decomposer is
a. Rabbit b. bacteria c. Tiger

7.The main sources of energy in an ecosystem is
a. Plant b. Sun c. water

8.The full form of AIDS is
a. Acquired Immuno Deficiency Syndrome
b. Acquired Immuno Diarrhoea Syndrome
c. All India development society

9.Which Greenhouse gas in the atmosphere absorb more radiated heat
a. Carbon dioxide b. water paper c. Methane

10.Which of the following is coldest region of atmosphere?
a. Exosphere b.Mesosphere c. Stratosphere

11.The normal greenhouse effect is essential for sustenance of life on earth as it helps in maintaining the average temperature of the earth to-
a. 15^0C b. 33^0C c.18^0C

12.The upper layer of the lithosphere is called
a. Mantle b. Core c. Crust

13.The main sources of urban atmospheric pollution is
a. Industries b. transportation c. domestic waste

14.The name of the biggest gaseous planet of solar system is
a. Mars b. Jupiter c. Saturn

15.The vector of the Malaria disease is
a. Flies b. Mosquito c. Rat

16.Number of biological Hotspot in the world is -
a.14 b.74 c.36

17.------------- is the indicator species of the environment
a. Lichen b. Dinosaur c. Mango tree

18.The writer of the origin of species
a. Charles Brown b. Charles Darwin c. Stephen Hawkings

19.The noise pollution measuring unit is
a.Kilogram b.calorie c.decibel

20.The Bhopal gas disaster happened in the year
a.1982 b.1983 c.1984

21. In which galaxy our solar system is located?
a. Milky Way b. Andromeda c.Ursa Major

22.The World Environment Day is celebrated on-
a.5th January b.5th June c.5th September

23.One of the main groundwater pollutants is
a. Mercury b. lead c. arsenic

Group- B

Answer any 6 questions from the following. 10x6=60

1. Mention the importance of biodiversity.What is mass extinction? Write about conservation biodiversity? 4+2+4

2. Define noise pollution? Discuss about the sources of noise pollution. Write the impacts of noise pollution. 2+4+4

3. Write any two short notes. 5x2=10
i) Food chain
ii) Indoor airpollution
iii) Ozone layer depletion
iv) Sources of air pollution.

4. What is environment? Write about the environmental components. How did oxygen evolve on primitive Earth?2+3+5

5. What is Ecosystem? What is ecological succession? Describe about pond ecosystem. 2+3+5

6. What is Global warming? Why it is called greenhouse effect? Discuss the environmental consequences of global warming. 2+3+5

7. Describe about the atmospheric layers with suitable diagram. 10

8. Women in Garhwal Himalayan region started a special movement by hugging the trees in the 1970s to protect their environment. What is the name of this movement? Write briefly about this movement. 1+9

Aliah University
End-Semester (spring) Practical Examination 2025
Semester – II (4 Year B.Sc. Major in Statistics)
Course Title: Descriptive Statistics
Course Code: STAUGMCC1103

Time: 2:00 Hours Full Marks: 20

Answer all the questions.

1. Consider the following frequency distribution:

Class Boundary	Frequency	Class Boundary	Frequency
2.5 – 7.5	12	32.5 – 37.5	176
7.5 – 12.5	28	37.5 – 42.5	120
12.5 – 17.5	65	42.5 – 47.5	66
17.5 – 22.5	121	47.5 – 52.5	27
22.5 – 27.7	175	52.5 – 57.5	9
27.5 – 32.5	198	57.5 – 62.5	3

i) Draw the histogram based on the above data.
ii) Calculate the mean, median and standard deviation of the above distribution. 3+(2+2+2)

2. Consider the following data, where X=marks in Mathematics and Y=Marks in Statistics.

X	59	65	45	52	60	62	70	55	45	49
Y	75	70	55	65	60	69	80	65	59	61

Compute the regression equation of Y on X: If a student gets 61 in mathematics, what would you estimate his/her marks in Statistics? 5+1

3. Viva-Voce 5

$y = a + bx$

$\sum y = Na + b\sum x$

$\sum xy = a + b\sum x^2$

Aliah University
End-Semester (Autumn) Practical Examination 2024
Semester – I (4 Year B.Sc. Honours in Statistics)
Course Title: Demography and Vital Statistics
Course Code: STAUGMCC1102
Time: 2:00 Hours Full Marks: 20

M.I $\Sigma(b_i - 100)$

Answer all the questions.

1. In the second and third column of the following table are given the age specific death rates for Poland and Sweden for the year 1957. The figures in the 4th column give the age-distribution of a standard population adopted by the International Statistical Institute (ISI).

Age	Death Rate (per thousand)		Number in ISI standard million
	Poland	Sweden	
0 – 4	18.870	4.348	119900
5 – 14	0.759	0.465	206900
15 – 24	1.385	0.767	183200
25 – 34	2.048	1.075	147900
35 – 44	3.326	1.882	120500
45 – 54	7.006	4.669	93900
55 – 64	18.111	12.477	70800
65 – 74	45.795	34.060	40500
75 and above	124.258	116.443	16400

Compute the standardized death rate for Poland and Sweden taking the ISI population as standard. 5

2. Complete the below life table. 5

Age x	l_x	d_x	q_x	L_x	T_x
10	74000		0.00350		
11			0.00338		
12			0.00361		
13			0.00420		
14			0.00517		
15			0.00530		
16			0.00538		
17			0.00544		
18			0.00549		
19			0.00554		2607040

3. Suppose in your data set, the population size at 20 to 70 is as follows:
Calculate Whipple's and Myre's index and comment.
35 30 35 36 32 36 51 51 53 48 80 48 44 47 50 55 44 73 46 47 47 53 44 52 49
53 52 70 64 35 57 44 44 58 58 57 49 53 54 52 44 48 59 69 61 51 56 46 54 71 33

4. Viva-Voce 5